AWS ETL Project: Social Media Data Processing using AWS Glue

1. Introduction

This project demonstrates the implementation of an ETL (Extract, Transform, Load) pipeline using cloud services. The goal is to process social media data (tweets and blogs), perform transformations such as joining datasets, extracting hashtags, and generating aggregated insights. The processed data is then stored for analysis.

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2. Objective

- To build an ETL pipeline using AWS services
- To extract data from Amazon S3
- To transform data using AWS Glue Visual ETL
- To load processed data into an S3 bucket
- To analyze data using SQL queries

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3. Tools and Technologies Used

- Amazon S3 (Storage)
- AWS Glue (ETL Processing)
- AWS IAM (Access Management)
- Amazon Athena (Querying)

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4. Architecture Overview

1. Data is uploaded into Amazon S3
2. AWS Glue Crawlers scan the data and create metadata tables
3. AWS Glue ETL job processes and transforms the data
4. Output is stored in another S3 bucket

5. Data is queried using SQL via Athena

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5. Implementation Steps

5.1 S3 Bucket Creation

- Created bucket: etl-twitter-blog
- Created folders:
    - etl-social-media
    - blog-data
- Uploaded datasets:
    - sample_tweets.csv
    - sample_blogs.csv
- Created output bucket: etl-cep-output

5.2 AWS Glue Database

- Database name: social_media_data

5.3 Classifiers

- twitter_data (CSV)
- blog_data (CSV)

5.4 Crawlers

- tweet-crawl → tweets data
- blog-crawler → blog data

Tables created:

- sample_tweets_csv

- sample_blogs_csv

5.5 IAM Role

- Role: glue-role
- Policy: AdministratorAccess

5.6 ETL Job (Visual ETL)

Transformations:

1. Join → user_id
2. Drop Fields → remove duplicate column
3. Regex Extractor → extract hashtags using #(\w+)
4. Aggregate:
   - Count of tweets
   - Minimum timestamp

5.7 Output

- Stored in S3 bucket: etl-cep-output

5.8 Query Execution

Used Athena to run SQL queries:

SELECT user_id, COUNT(tweet_id)

FROM sample_tweets_csv

GROUP BY user_id;

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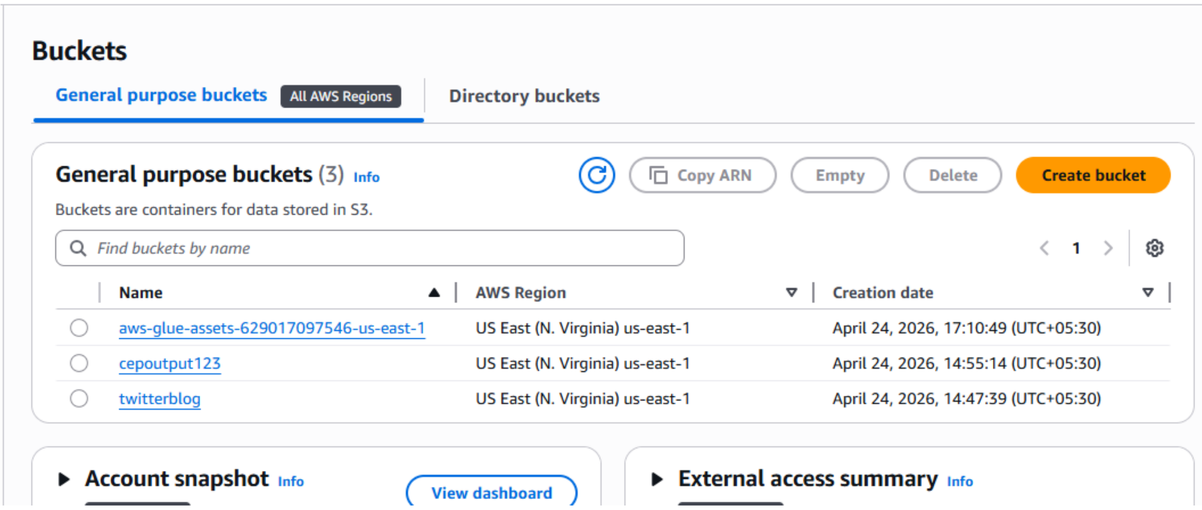
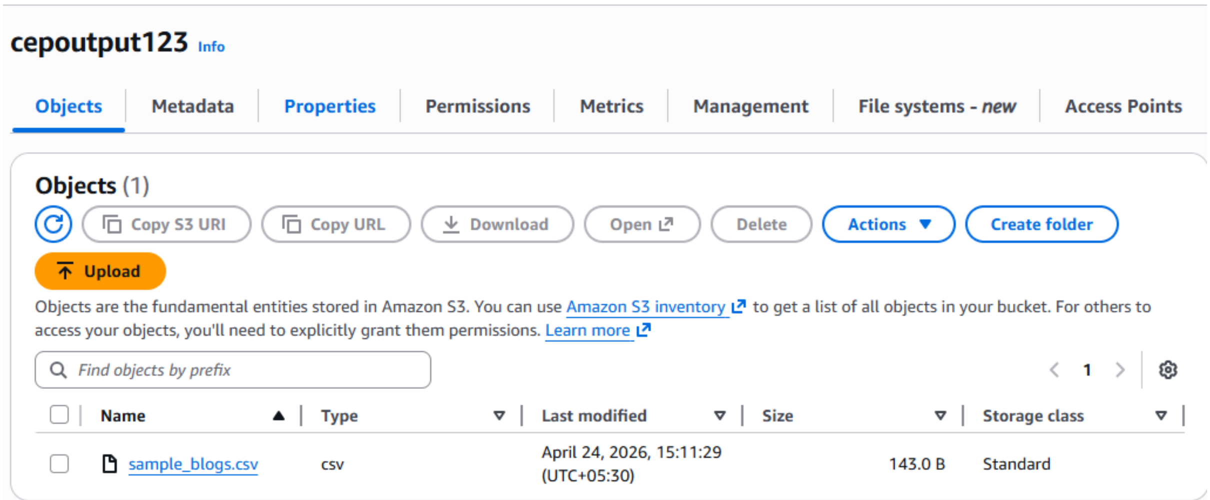
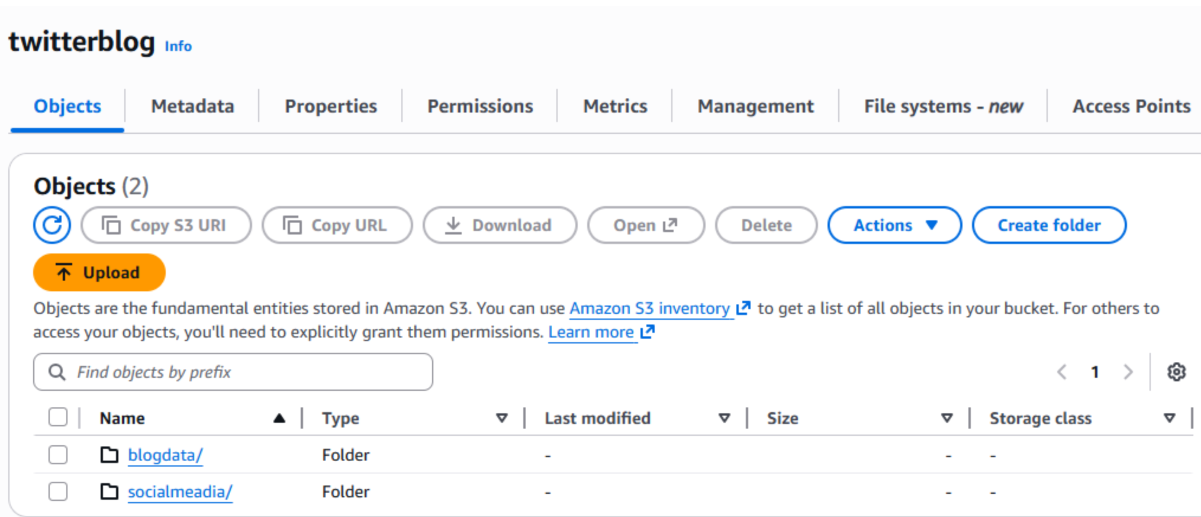
6. Screenshots

6.1 S3 Bucket Creation

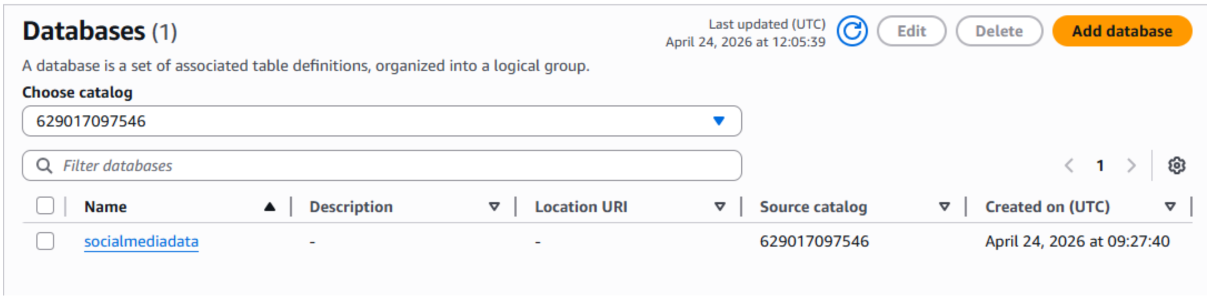


6.2 Folder Creation and File Upload

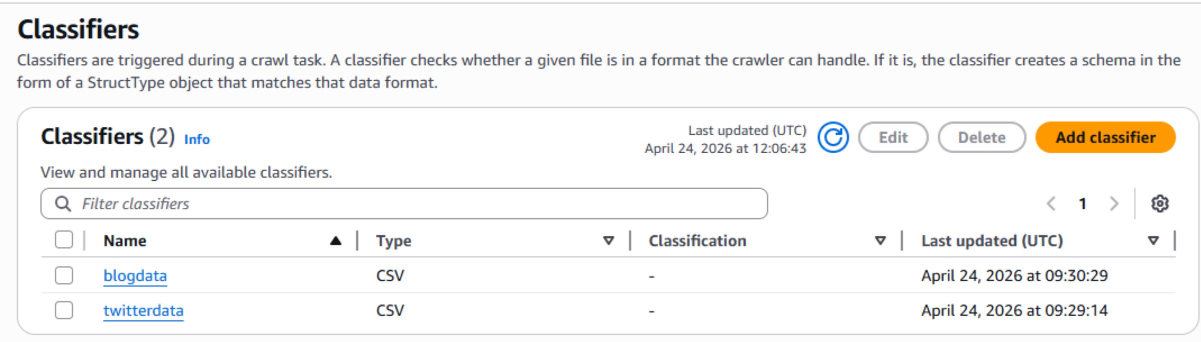




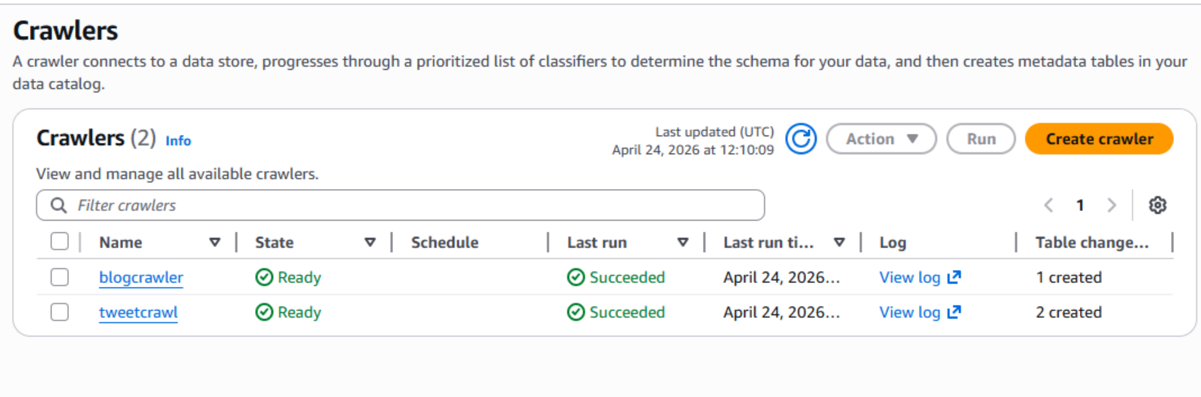
6.3 Glue Database Creation



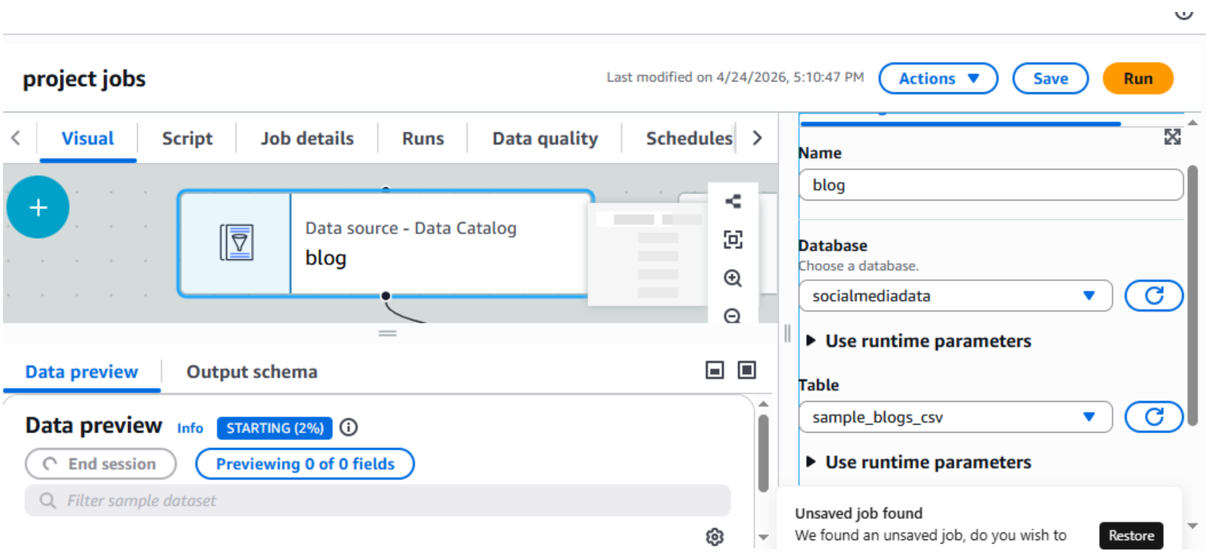
6.4 Classifiers Setup



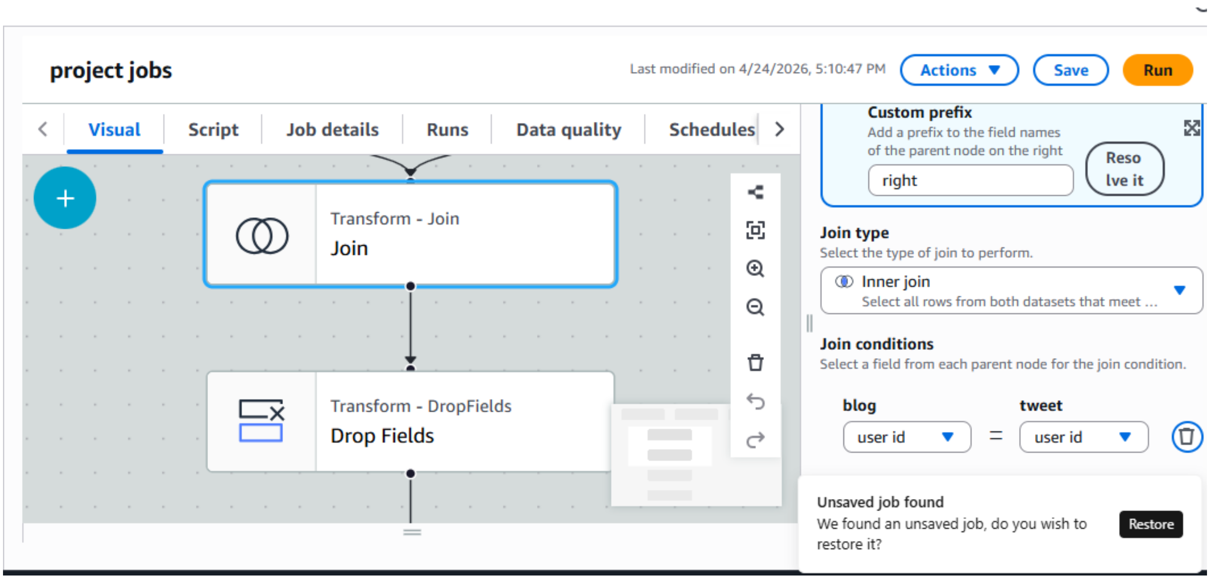
6.5 Crawlers Configuration and Run



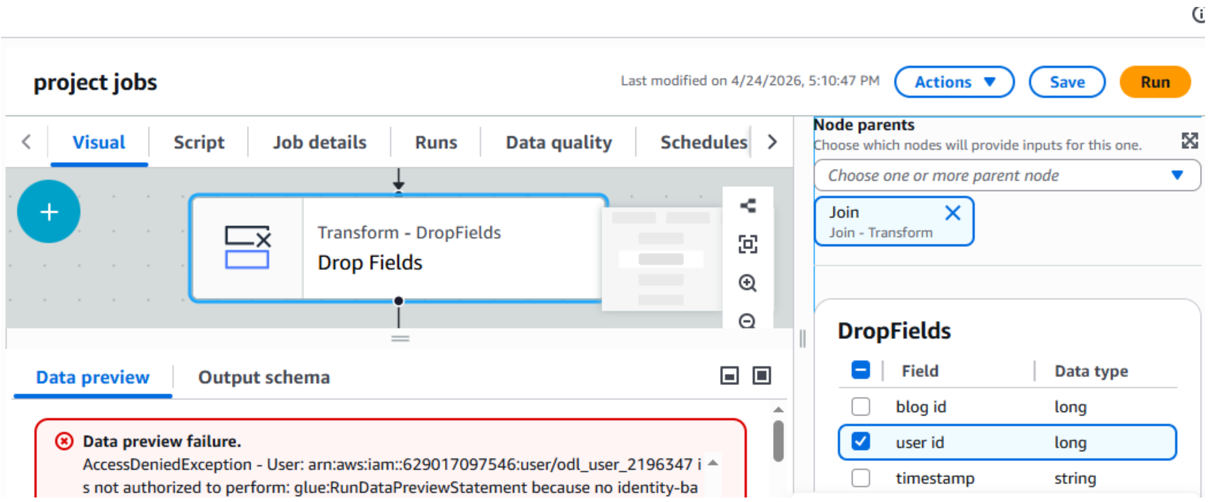
6.6 Data Catalog Tables



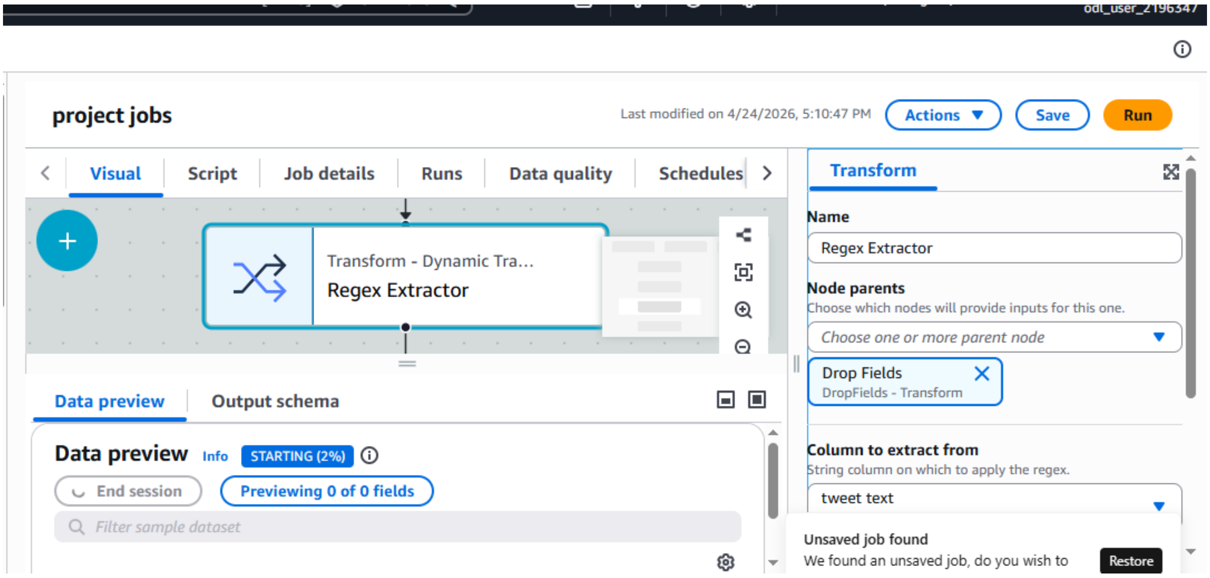
Join Configuration



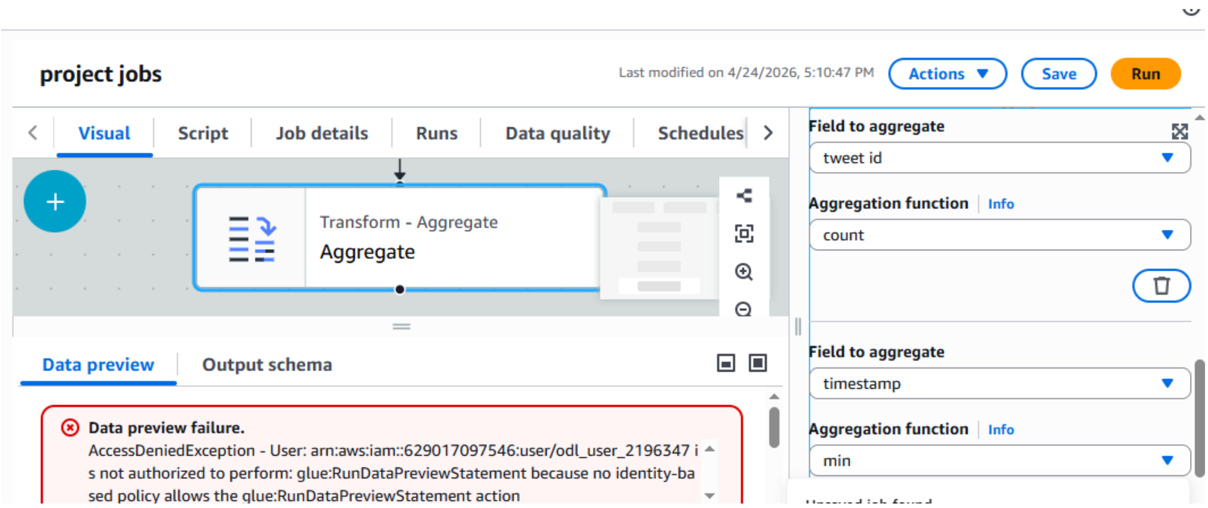
Drop filled Configuration

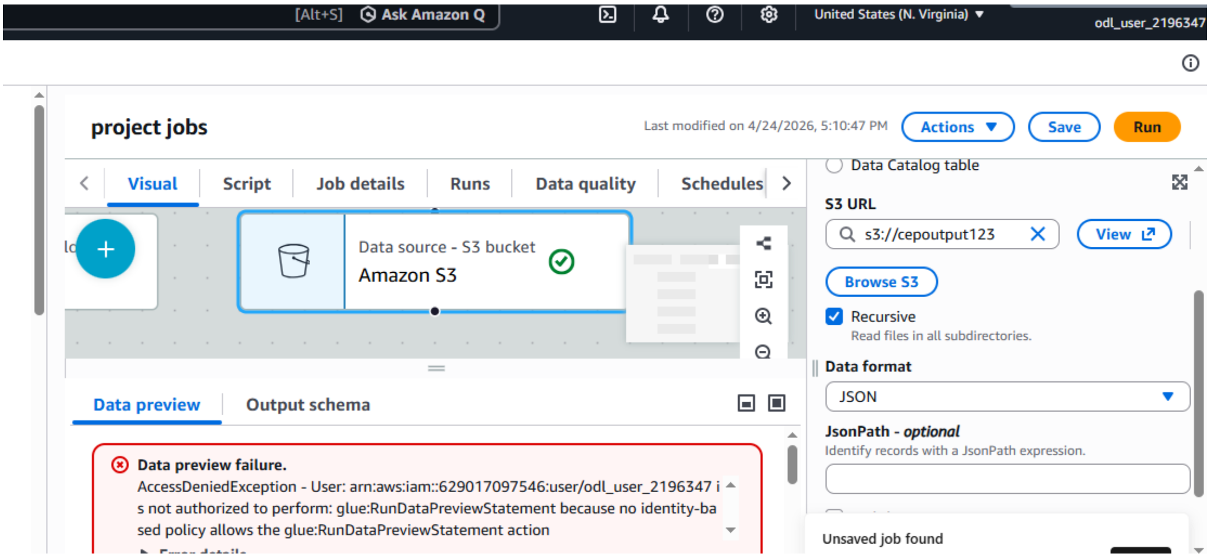


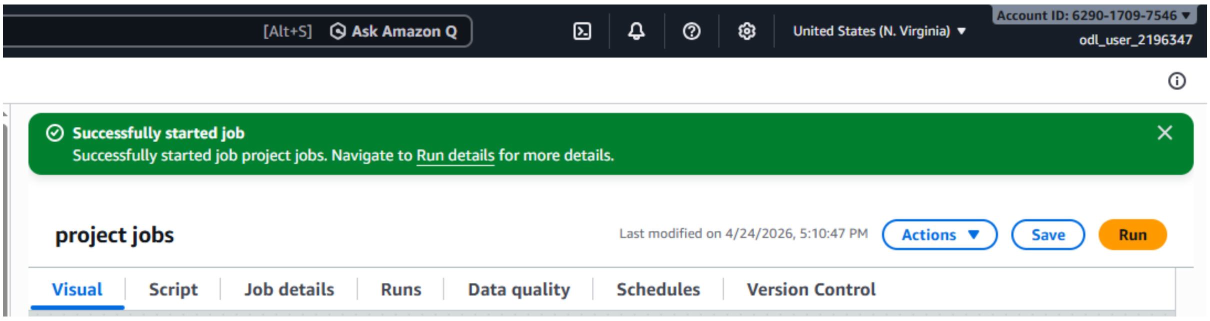
6.9 Regex Extractor Configuration



6.10 Aggregate Configuration







7. Results

- Successfully processed and joined datasets
- Extracted hashtags from tweets
- Generated aggregated insights
- Stored results in S3

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8. Challenges Faced

- Issues connecting nodes in Visual ETL
- S3 permission errors
- UI glitches in Glue

9. Conclusion

The project successfully demonstrates building a cloud-based ETL pipeline using AWS services. It highlights efficient data processing, transformation, and storage.

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10. Future Enhancements

- Real-time data streaming
- Dashboard visualization
- Automation using scheduling
